PLRD: Partially Linear Regression Discontinuity Inference

Aditya Ghosh Guido Imbens Stefan Wager

Stanford University

What is a Regression Discontinuity Design (RDD)?

- Running variable X decides treatment: $W = \mathbf{1}(X \geq c)$. Outcome $Y = Y(W)$.
- We want to estimate and draw inference on the conditional average treatment effect at the threshold:

$$\tau_c := \mu_1(c+) - \mu_0(c-), \quad \text{where} \quad \mu_w(x) = \mathbb{E}[Y|X = x, W = w].$$

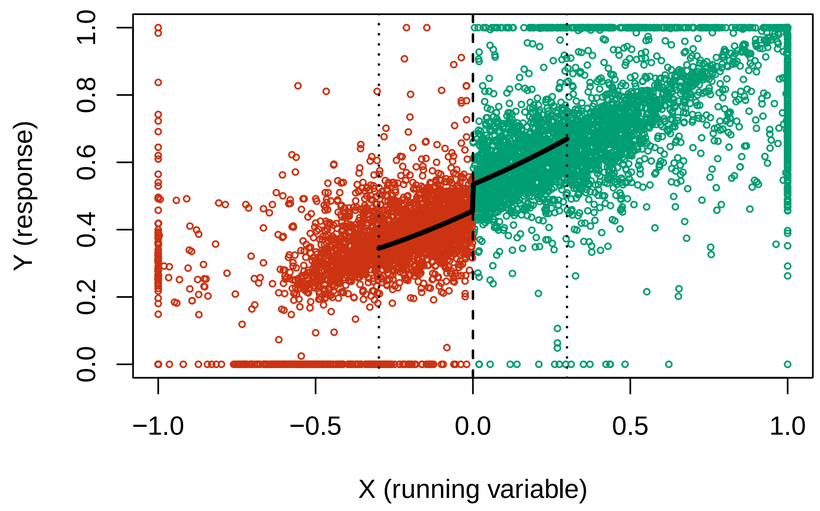

Figure 1. Incumbency advantage in U.S. house elections (Lee, 2008)

Another example: studying the effect of summer school on students' test scores when eligibility is threshold-based (Jacob and Lefgren, 2004; Matsudaira, 2008).

Existing methods for inference

$$(\widehat{\tau}_{\text{LLR}}, \widehat{\alpha}, \widehat{\beta}) = \underset{\tau,\alpha,\beta}{\text{argmin}} \sum_{i=1}^{n} K\left(\frac{X_i - c}{h}\right)(Y_i - \alpha - \tau W_i - \beta_0(X_i - c)_- - \beta_1(X_i - c)_+)^2$$

Traditional local linear regression methods in RDD often struggle with bias-variance trade-off (MSE optimal bandwidth sets bias and s.d. of the same order), which complicates inference. Two popular approaches for inference are:

- **Bias-corrected CIs:** estimate the bias using higher-order polynomial regression (see, e.g., Calonico et al. (2014, 2015, 2017, 2019)). (R package: **rdrobust**)
  Advantage: Narrow CIs. Downside: May fail to deliver nominal coverage!
- **Bias-aware CIs:** inflate the confidence intervals to account for worst-case bias under a function class for the functions $\mu_w(x)$ (see, e.g., Armstrong and Kolesar (2018, 2020); Imbens and Wager (2019)). (candidate R package: **RDHonest**)
  Advantage: More transparent. Downside: Often too conservative in practice.

Our contribution. We combine higher-order smoothness (on par with the bias-correction literature) with a semiparametric assumption to achieve transparent, asymptotically valid, bias-aware inference with narrow CIs.

Empirical comparison using Generative AI

	rdrobust (b.c.)		rdrobust (rob.)		RDHonest		plrd	
Dataset	EC	width	EC	width	EC	width	EC	width
Lee (2008)	91.0%	0.036	95.4%	0.042	97.6%	0.052	98.3%	0.035
Cattaneo et al. (2015)	90.7%	5.353	95.1%	6.281	97.4%	6.980	97.3%	4.525
Matsudaira (2008) (read.)	93.8%	0.061	96.9%	0.069	98.4%	0.096	97.1%	0.060
Matsudaira (2008) (math)	88.6%	0.045	93.0%	0.054	97.8%	0.072	97.0%	0.051
Jacob and Lefgren (2004) (read.)	88.4%	0.067	95.0%	0.080	98.2%	0.104	96.5%	0.065
Jacob and Lefgren (2004) (math)	87.8%	0.078	93.4%	0.093	97.1%	0.120	95.0%	0.069
Oreopoulos (2006)	84.1%	0.196	94.6%	0.264	98.9%	0.316	96.1%	0.194
Ludwig and Miller (2007)	91.2%	4.058	95.7%	4.786	97.6%	5.238	97.4%	3.542
Meyersson (2014)	89.5%	6.362	93.3%	7.531	96.3%	8.099	97.3%	5.095

Table 1. Empirical coverage (EC) and width of 95% CIs from different methods, on simulated RDDs calibrated to nine applications. We highlight (in green) the entries whose coverage is not significantly below nominal 95% coverage and (in blue) the entries with the shortest widths among those delivering valid coverage.

Following Athey et al. (2024) we generate simulation distributions by training a Wasserstein Generative Adversarial Neural Network (WGAN) on each of the original datasets (see Fig. 2 below for an illustration).

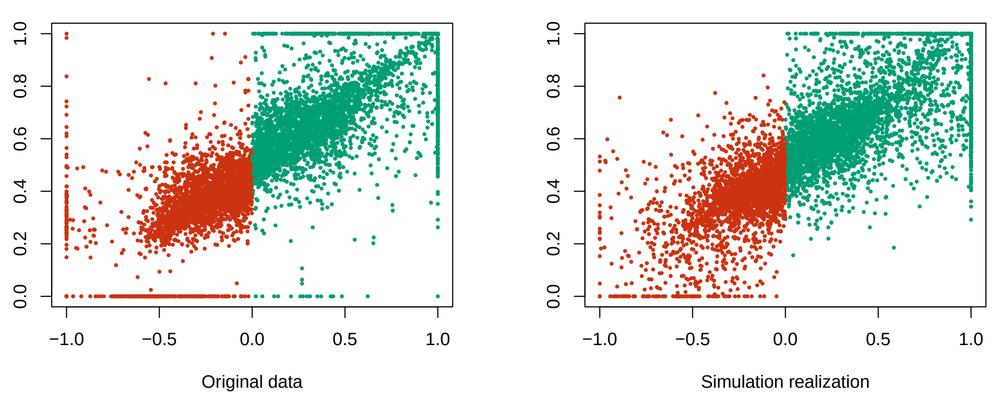

Figure 2. The original data from one of our benchmark datasets, Lee (2008), as well as a simulated dataset produced using a WGAN trained to the original data.

Minimax linear estimation and bias-aware inference

- The LLR estimator $\widehat{\tau}_{\text{LLR}}$ takes the form $\widehat{\tau}(\gamma) = \sum_{i=1}^{n} \gamma_i Y_i$ for some weights γ_i.
- We can use convex optimization to directly find weights γ_i that minimize worst-case MSE (conditional on X_i's). More precisely, we solve the minimax problem:

$$\widehat{\gamma} := \underset{\gamma}{\text{argmin}} \underbrace{\sup_{\mu_0, \mu_1 \in \mathcal{M}} \left(\sum_{i=1}^{n} \gamma_i \mu_{W_i}(X_i) - (\mu_1(c) - \mu_0(c))\right)^2}_{\text{worst-case squared bias over } \mathcal{M}} + \underbrace{\sigma^2 \|\gamma\|_2^2}_{\text{variance proxy}}. \quad (1)$$

  where $\mathcal{M}$ is a function class for the regression functions $\mu_w(x)$.
- As a by-product of this optimization procedure, we obtain an upper bound $\widehat{b}$ on the bias of the resulting estimator $\widehat{\tau}(\widehat{\gamma})$.
- With the bound $\widehat{b}$ on the conditional bias, we follow Imbens and Manski (2004) to construct an asymptotically valid bias-aware confidence interval.

Our PLRD method

We operate under the following assumptions:

- (smoothness) The regression functions $\mu_0(x)$ and $\mu_1(x)$ are three times continuously differentiable in x.
- (partially linear model) The conditional average treatment effect function $\tau(x) = \mu_1(x) - \mu_0(x)$ depends linearly on x, i.e., $\tau(x) = \tau_c + \beta(x - c)$ for some $\beta \in \mathbb{R}$.

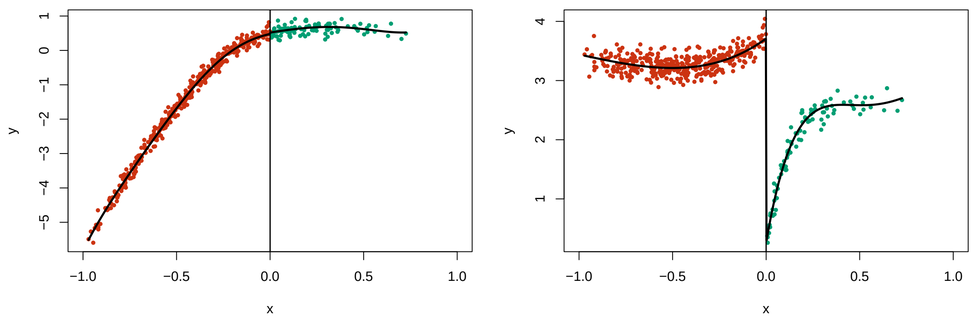

Figure 3. Two simulation settings from Calonico et al. (2014). Our partially linear assumption is plausible for the data on the left panel but not for the right panel.

The following steps summarize our PLRD method:

- The second derivatives $\mu_w''(x)$ are B-Lipschitz in a neighborhood of c. We estimate B using global cubic regressions on either side, with data-splitting.
- We form the set $\mathcal{M}_{\widehat{B}}$ of all possible choices of $\mu_0(x)$ and $\mu_1(x)$ consistent with the $\widehat{B}$-Lipschitz curvature bound and satisfying the partially linear assumption.
- We find the minimax linear estimator within this class $\mathcal{M}_{\widehat{B}}$ using convex duality and numerical optimization. Thus, our point estimator is given by

$$\widehat{\tau}_{\text{PLRD}} := \sum_{i=1}^{n} \widehat{\gamma}_i Y_i,$$

  where $\widehat{\gamma}$ solves (1) with $\mathcal{M} = \mathcal{M}_{\widehat{B}}$.
- Finally, we construct a bias-aware confidence interval $\mathcal{I}_\alpha$ for τ_c as follows:

$$\mathcal{I}_\alpha := [\widehat{\tau}_{\text{PLRD}} \pm \widehat{h}_\alpha], \quad \widehat{h}_\alpha := \min\{h : \mathbb{P}(|b + \widehat{s}(\widehat{\gamma})Z| \leq h) \geq 1 - \alpha \text{ for all } |b| \leq \widehat{b}\},$$

  where $Z \sim \mathcal{N}(0, 1)$, $\widehat{s}(\widehat{\gamma})$ is an estimate of the standard error, and $\widehat{b}$ is an upper bound on the bias of $\widehat{\tau}_{\text{PLRD}}$ that we obtain as a by-product of convex duality.

Our method is fully data-driven and implemented in the R package `plrd`.

Theoretical guarantees

We derive asymptotic properties of $\widehat{\tau}_{\text{PLRD}}$ using **convex duality** and **empirical process theory**. In particular, we show that under regularity conditions,

$$(\widehat{\tau}_{\text{PLRD}} - b(\widehat{\gamma}) - \tau_c) \,/\, s(\widehat{\gamma}) \xrightarrow{d} \mathcal{N}(0, 1),$$

where $s^2(\widehat{\gamma}) = \sum_{i=1}^{n} \widehat{\gamma}_i^2 \sigma_i^2 = O_p(n^{-6/7})$, and $b(\widehat{\gamma}) = \sum_{i=1}^{n} \widehat{\gamma}_i \mu_{W_i}(X_i) - \tau_c$ is the conditional bias. Furthermore,

$$\limsup_{n \to \infty} \left(|b(\widehat{\gamma})| - \widehat{b}\right) \,/\, s(\widehat{\gamma}) \leq_p 0,$$

i.e., the learned bias bound $\widehat{b}$ is conservative for the true bias. As a consequence, the confidence intervals we construct are asymptotically valid.

Our proof strategy hinges on establishing fast-enough convergence of the empirical dual minimizer to the population-level dual minimizer that enables us to derive large-sample properties of $\widehat{\tau}_{\text{PLRD}}$ from its dual counterpart.

	rdrobust (b.c.)		rdrobust (rob.)		RDHonest		plrd	
	coverage	width	coverage	width	coverage	width	coverage	width
Setting 1	86.3%	0.207	91.5%	0.244	95.2%	0.290	94.6%	0.235
Setting 2	91.1%	0.313	93.2%	0.342	96.9%	0.436	51.5%	0.309
Setting 3	88.6%	0.218	93.1%	0.250	95.7%	0.301	95.9%	0.239
Setting 4	87.8%	0.207	93.0%	0.246	96.8%	0.275	97.5%	0.236

Table 2. Empirical coverage and average width of 95% CIs produced by different methods for the simulation settings in Calonico et al. (2014) and Imbens and Kalyanaraman (2012).

Email: ghoshadi@stanford.edu
arXiv preprint: 2503.09907
Website: ghoshadi.github.io